Lab 1: Packet analysis at application layer using Wireshark

SCSR1213 Network Communications

Universiti Teknologi Malaysia

Objective:

1. Understanding of network protocols by observing the sequence of messages exchanged between two protocol entities, delving down into the details of protocol operation, and causing protocols to perform certain actions and then observing these actions and their consequences.

2. To introduce students with the Wireshark software tool for packet analyzer.

3. To analyze protocol used in application layers such as http and dns.

Reference material: Computer Networking: A Top-Down Approach, 7th ed., J.F. Kurose and K.W. Ross.

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Section : 12

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PART A: Wireshark Getting Started

1.0 Introduction

The basic tool for observing the messages exchanged between executing protocol entities is called a **packet sniffer**. As the name suggests, a packet sniffer captures ("sniffs") messages being sent/received from/by your computer; it will also typically store and/or display the contents of the various protocol fields in these captured messages. A packet sniffer itself is passive. It observes messages being sent and received by applications and protocols running on your computer, but never sends packets itself. Similarly, received packets are never explicitly addressed to the packet sniffer. Instead, a packet sniffer receives a *copy* of packets that are sent/received from/by application and protocols executing on your machine.

Figure A.1 shows the structure of a packet sniffer. At the right of Figure 1 are the protocols (in this case, Internet protocols) and applications (such as a web browser or ftp client) that normally run on your computer. The packet sniffer, shown within the dashed rectangle in Figure A.1 is an addition to the usual software in your computer, and consists of two parts. The **packet capture library** receives a copy of every link-layer frame that is sent from or received by your computer. In Figure A.1, the assumed physical media is an Ethernet, and so all upper-layer protocols are eventually encapsulated within an Ethernet frame. Capturing all link-layer frames thus gives you all messages sent/received from/by all protocols and applications executing in your computer.

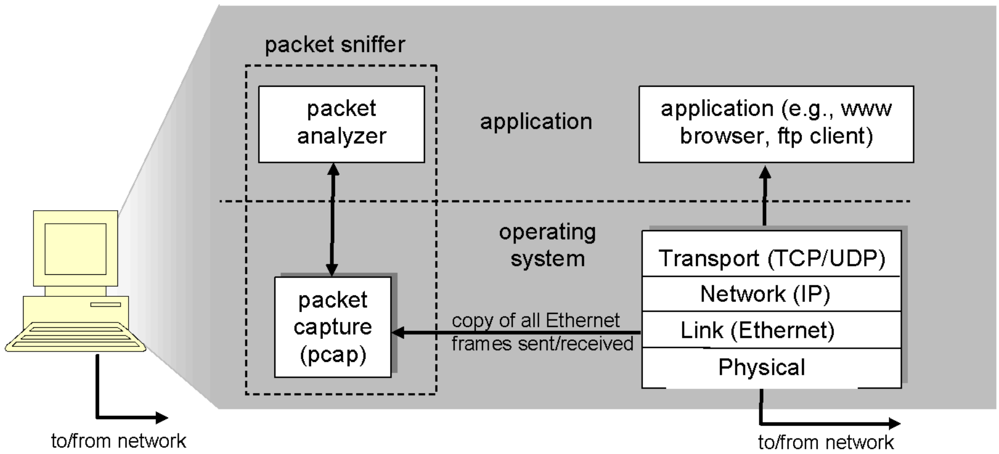


Figure A.1: Packet sniffer structure

The second component of a packet sniffer is the **packet analyzer**, which displays the contents of all fields within a protocol message. In order to do so, the packet analyzer must "understand" the structure of all messages exchanged by protocols. The packet analyzer understands the format of Ethernet frames, and so can identify the IP datagram within an Ethernet frame. It also understands the IP datagram format, so that it can extract the TCP segment within the IP datagram. Finally, it understands the TCP segment structure, so it can extract the HTTP message contained in the TCP segment. Finally, it understands the HTTP protocol and so, for example, knows that the first bytes of an HTTP message will contain the string "GET," "POST," or "HEAD".

2.0 Getting Wireshark Ready

- Download and install the Wireshark software
- Run Wireshark. Wireshark startup screen shown in Figure A.2.

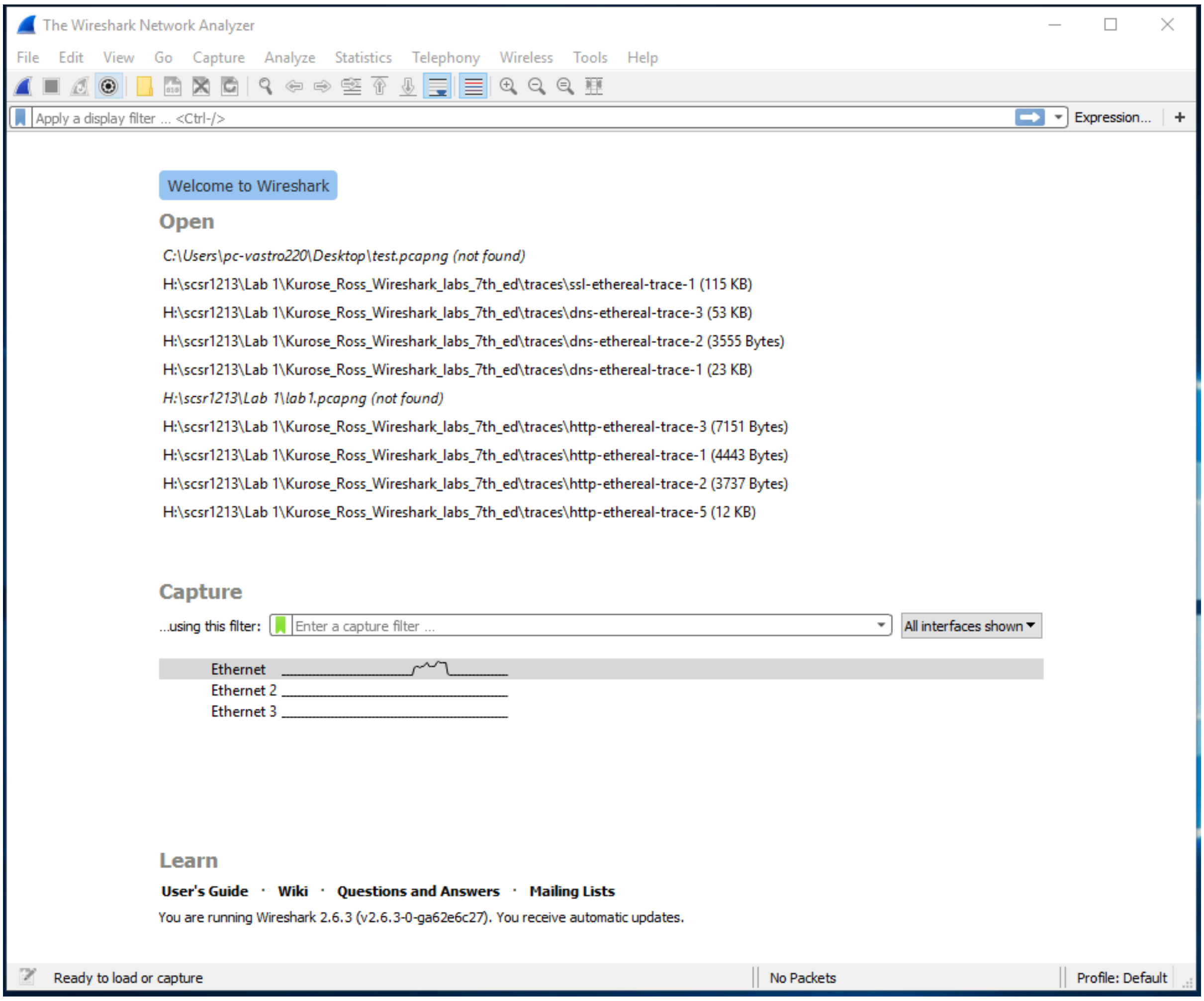


Figure A.2: Initial Wireshark startup screen

- The Wireshark interface has five major components as shown in Figure A.3.

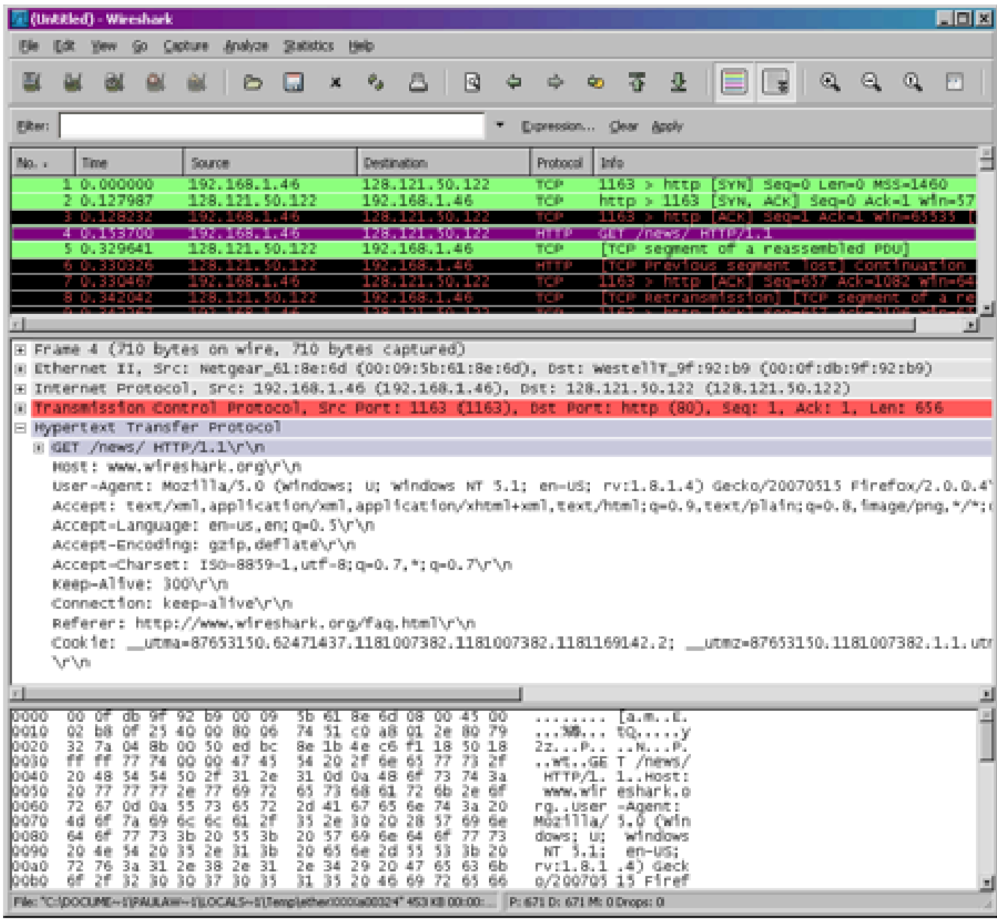


Figure A.3: Wireshark Graphical User Interface, during packet capture and

- The **command menus** are standard pulldown menus located at the top of the window.
- The **packet display filter field,** into which a protocol name or other information can be entered in order to filter the information displayed in the packet-listing window.
- The **packet-listing window** displays a one-line summary for each packet captured, including the packet number, the time at which the packet was captured, the packet's source and destination addresses, the protocol type, and protocol-specific information contained in the packet.
- The **packet-header details window** provides details about the packet selected (highlighted) in the packet-listing window. These details include information about the Ethernet frame and IP datagram that contains this packet. The amount of Ethernet and IP-layer detail displayed can be expanded or minimized by clicking on the plus minus boxes to the left of the Ethernet frame or IP datagram line in the packet details window. If the packet has been carried over TCP or UDP, TCP or UDP details will also be displayed, which can similarly be expanded or minimized. Finally, details about the highest-level protocol that sent or received this packet are also provided.
- The **packet-contents window** displays the entire contents of the captured frame, in both ASCII and hexadecimal format.

3.0 Test Run Wireshark

- Start up the Wireshark software.
- To begin packet capture, select the Capture pull down menu and pick Options menu. Select appropriate interfaces on your compute and click Start button to begin packet capture. Refer to Figure A.4

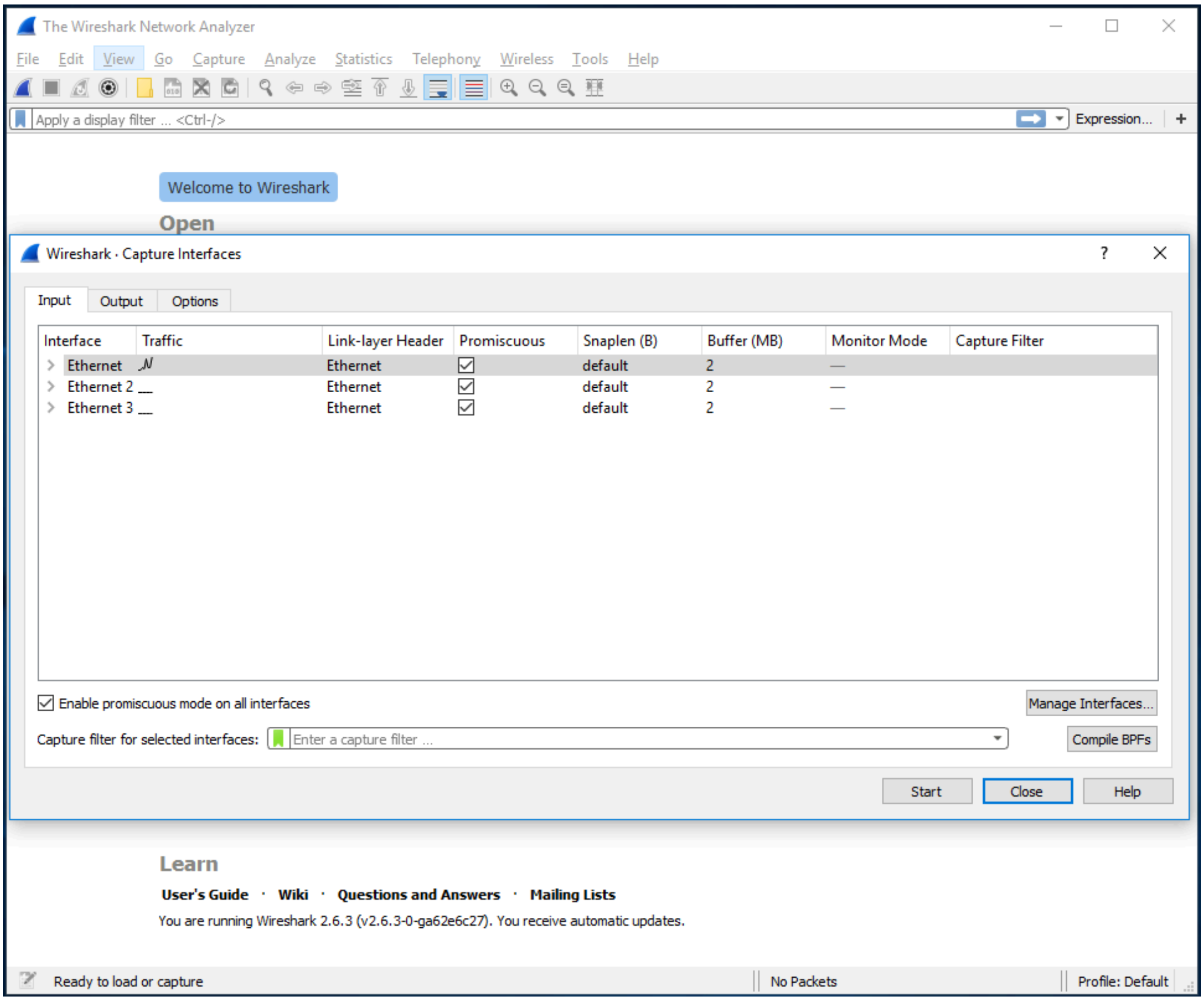


Figure A.4: Capture and Options Menu

- Once you begin packet capture, result will be shown as in Figure A.5.

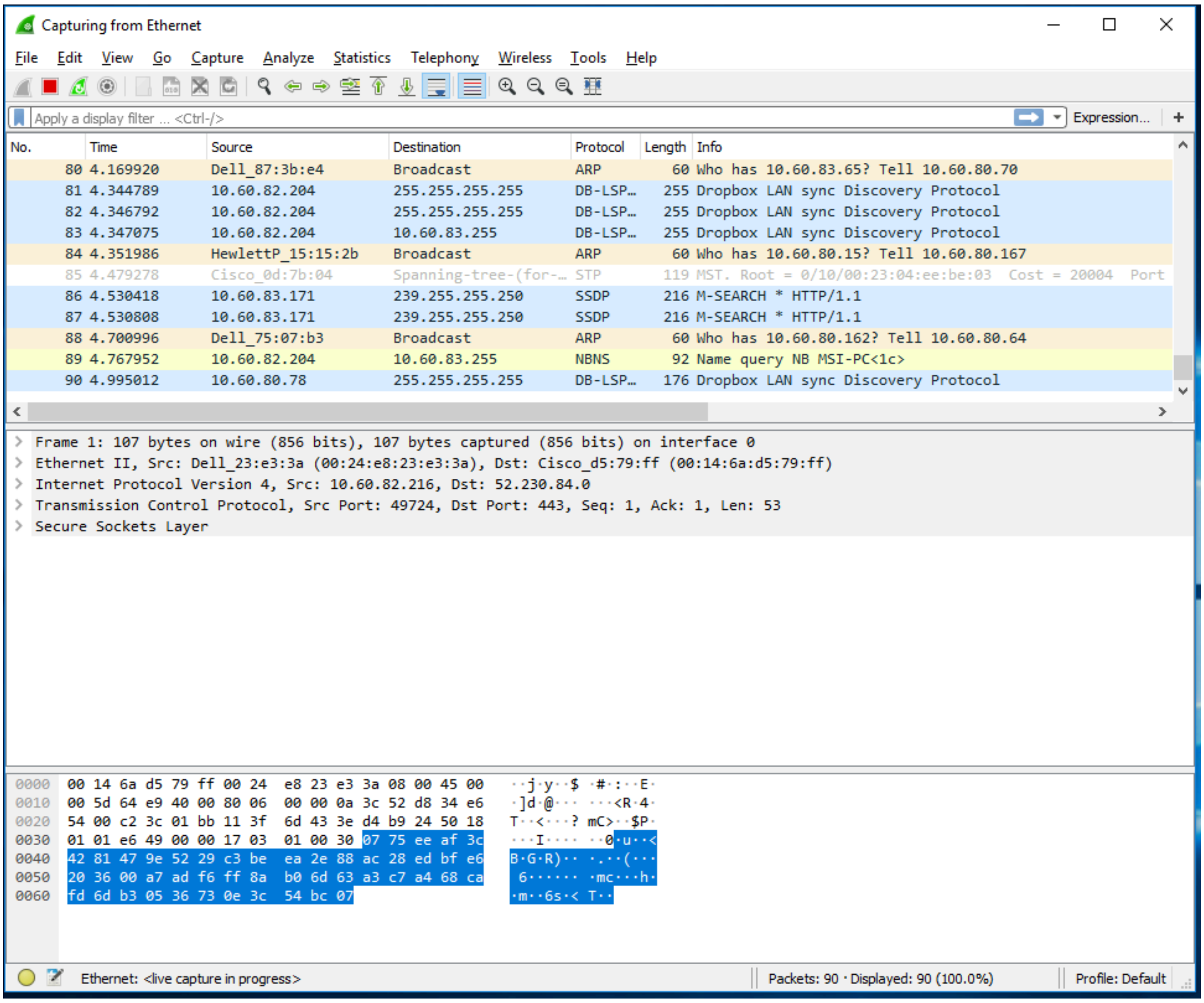


Figure A.5: Wireshark packet capture result

- By selecting Capture pulldown menu and selecting Stop, you can stop packet capture.

- Type “arp” in packet display filter field and press Enter key. This will cause only ARP message to be displayed in the packet-listing window as shown in Figure A.6.

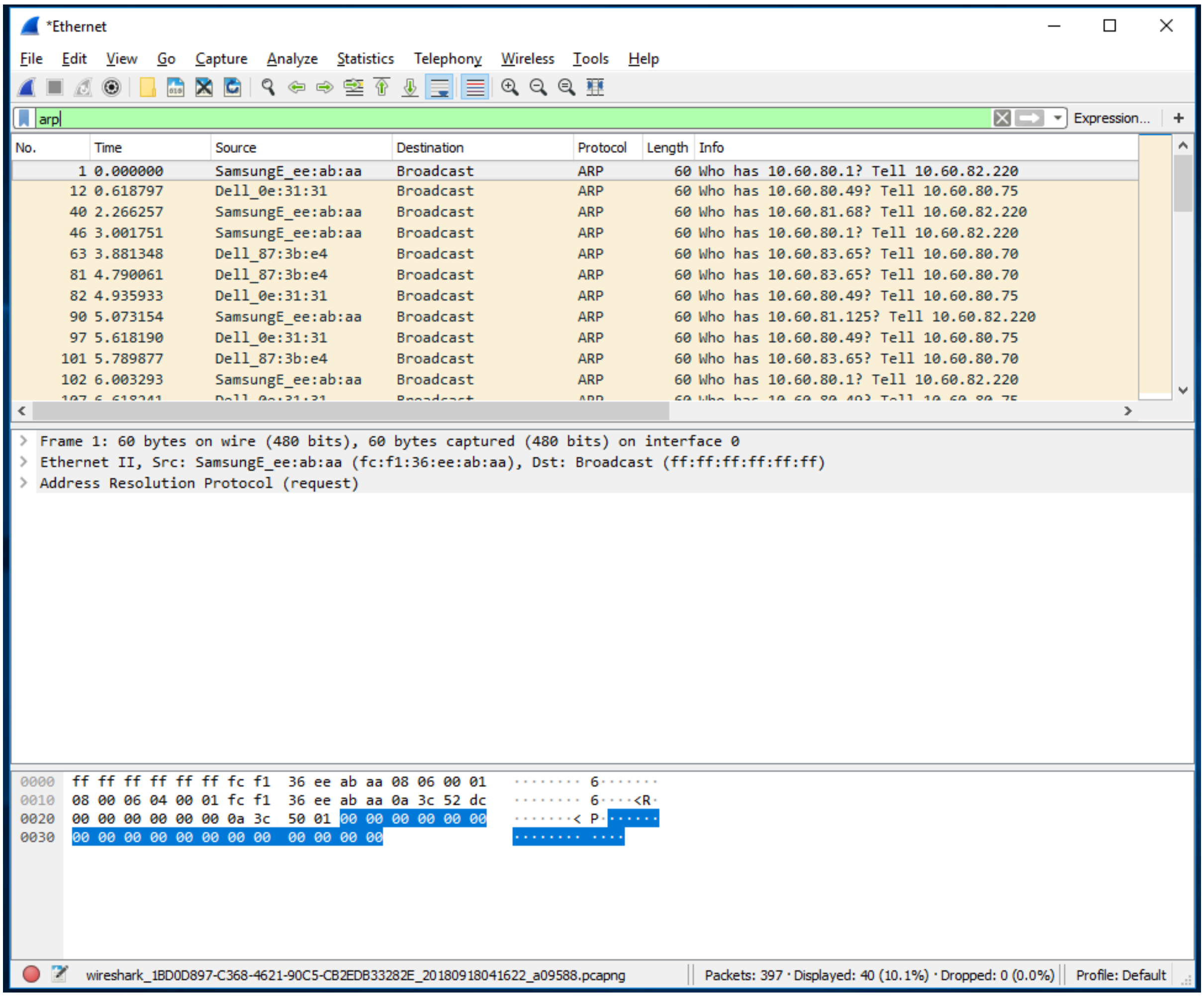


Figure A.6: ARP packet capture

- To save the trace result, use File pulldown menu and select Save function as shown in Figure A.7.

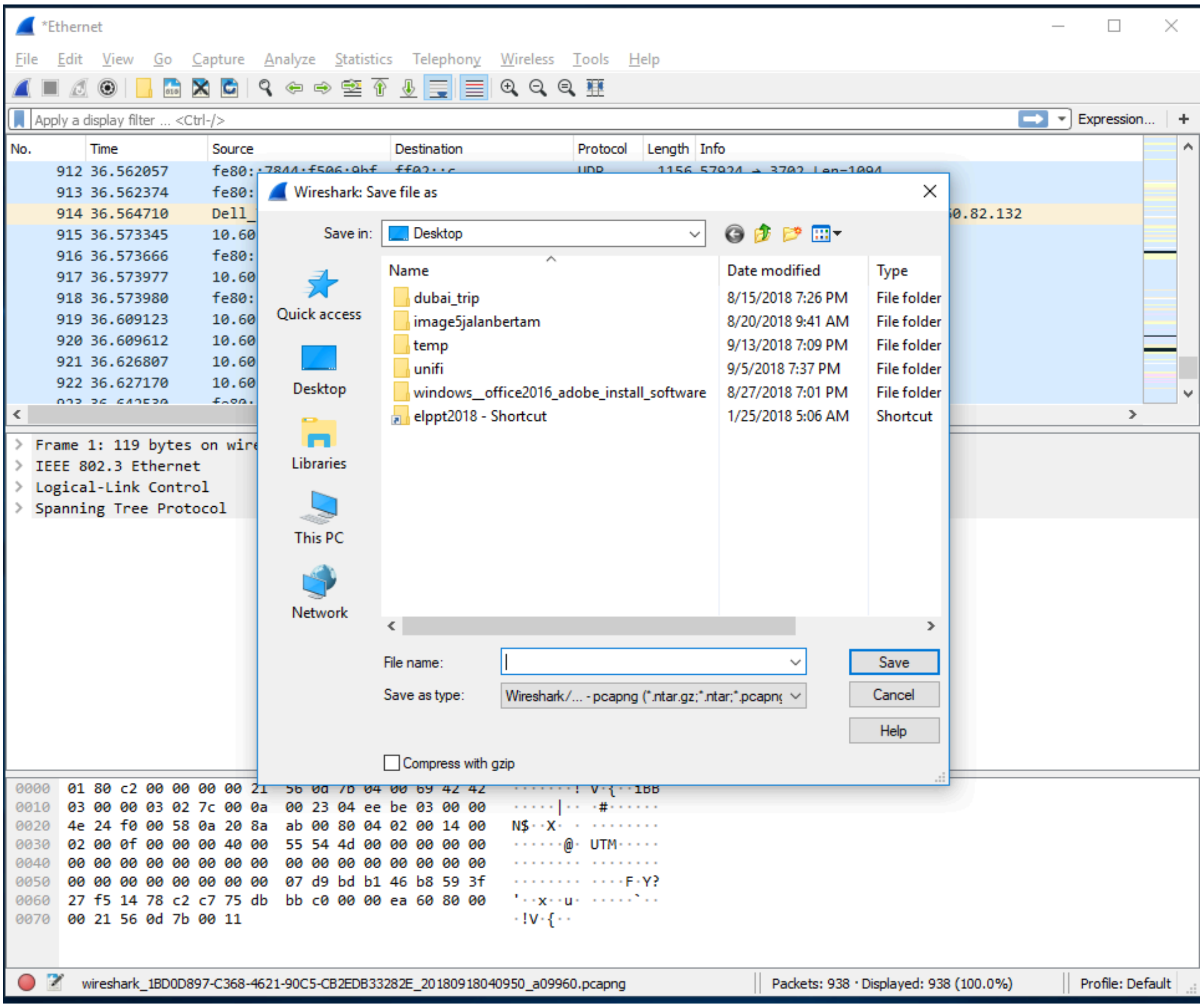


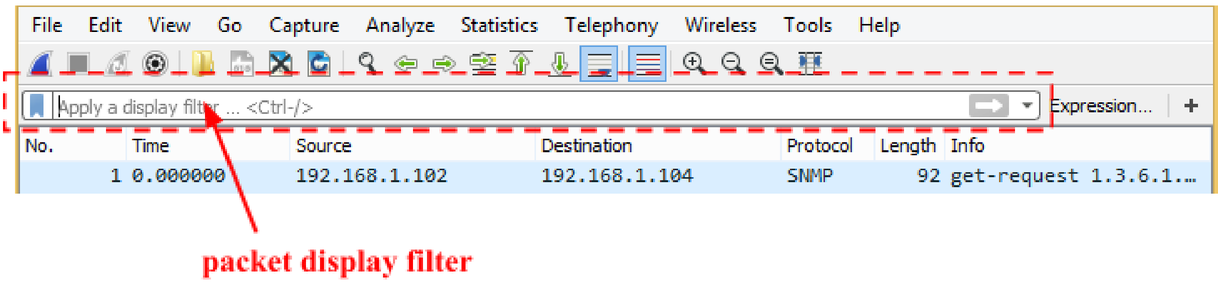
Figure A.7: Save Wireshark trace result

PART B: HTTP Trace

In this part, we'll explore several aspects of the HTTP protocol: the basic GET/response interaction, HTTP message formats and retrieving HTML files with embedded objects. Before beginning these labs, you might want to review Section 2.2 of the textbook.

B.1 The Basic HTTP GET/response interaction

- Open packet trace file **lab1-http-B01.pcapng**.
- Enter "**http**" (just the letters, not the quotation marks) in the **packet display filter field**, so that only captured HTTP messages will be displayed later in the packet-listing window. Refer to figure below:



- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. What version of HTTP is the server running?
   Ans: HTTP version 1.1

2. What is the IP address of the client computer?
   Ans: 192.168.1.102

3. What is the IP address of the gaia.cs.umass.edu server?
   Ans: 128.119.245.12

4. How many bytes of content are being returned to client browser?

   Ans: 439 bytes.

5. What is the status code returned from the server to client browser?

   Ans: 200 OK and 404 Not Found.

B.2 The HTTP CONDITIONAL GET/response interaction

- Open packet trace file **lab1-http-B02.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE" line in the HTTP GET?

   Ans: No.

2. Inspect the contents of the server response after the first GET request from client. Did the server explicitly return the contents of the file? How can you tell?

   Ans: Yes, in line 10, the response has status code 200 OK(text/html).

No.	Time	Source	Destination	Protocol	Length	Info
8	2.331268	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-2.html HTTP/1.1
10	2.357902	128.119.245.12	192.168.1.102	HTTP	739	HTTP/1.1 200 OK (text/html)
14	5.517390	192.168.1.102	128.119.245.12	HTTP	668	GET /ethereal-labs/lab2-2.html HTTP/1.1
15	5.540216	128.119.245.12	192.168.1.102	HTTP	243	HTTP/1.1 304 Not Modified

3. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE:" line in the HTTP GET? If so, what information follows the "IF-MODIFIED-SINCE:" header?

   Ans: Yes. If-Modified-Since: Tue, 23 Sep 2003 05:35:00 GMT\r\n

```
Keep-Alive: 300\r\n
Connection: keep-alive\r\n
If-Modified-Since: Tue, 23 Sep 2003 05:35:00 GMT\r\n
If-None-Match: "1bfef-173-8f4ae900"\r\n
```

4. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.

   Ans: The status code is 304 Not Modified. The server does not specifically return the contents of the file. 304 Not Modified means that the requested resource has not changed since the last request. The server tells the browser to use the cached version instead of downloading it again.

14	5.517390	192.168.1.102	128.119.245.12	HTTP	668	GET /ethereal-labs/lab2-2.html HTTP/1.1
15	5.540216	128.119.245.12	192.168.1.102	HTTP	243	HTTP/1.1 304 Not Modified

B.3 HTML Documents with Embedded Objects

- Open packet trace file **lab1-http-B03.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. How many HTTP GET request messages did client browser send?
Ans: 3.

	Time	Source	Destination	Protocol	Length	Info
10	7.236929	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-4.html HTTP/1.1
12	7.260813	128.119.245.12	192.168.1.102	HTTP	1057	HTTP/1.1 200 OK (text/html)
17	7.305485	192.168.1.102	165.193.123.218	HTTP	625	GET /catalog/images/pearson-logo-footer.gif HTTP/1.1
20	7.308803	192.168.1.102	134.241.6.82	HTTP	609	GET /~kurose/cover.jpg HTTP/1.1
25	7.333054	165.193.123.218	192.168.1.102	HTTP	912	HTTP/1.1 200 OK (GIF89a)
54	7.589877	134.241.6.82	192.168.1.102	HTTP	1096	HTTP/1.0 200 Document follows (JPEG JFIF image)

2. To which Internet addresses were these GET requests sent?

Ans:
- 128.119.245.12
- 165.193.123.218
- 134.241.8.82

3. How many bytes of content are being returned to client browser for the **pearson-logo-footer.gif** image file?
Ans: 912 bytes.

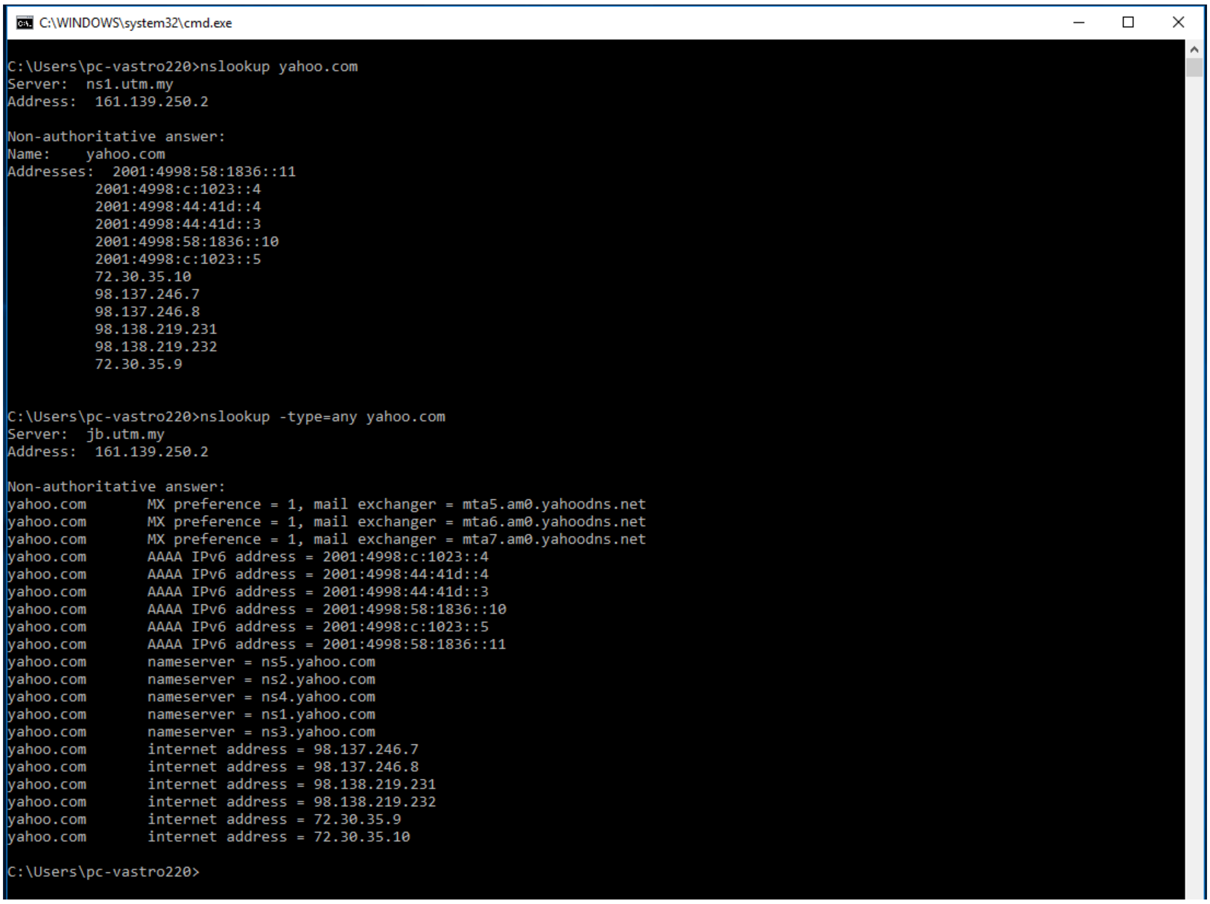
4. How many bytes of content are being returned to client browser for the **cover.jpg** image file?
Ans: 1096 bytes.

PART C: DNS Trace

1.0 nslookup

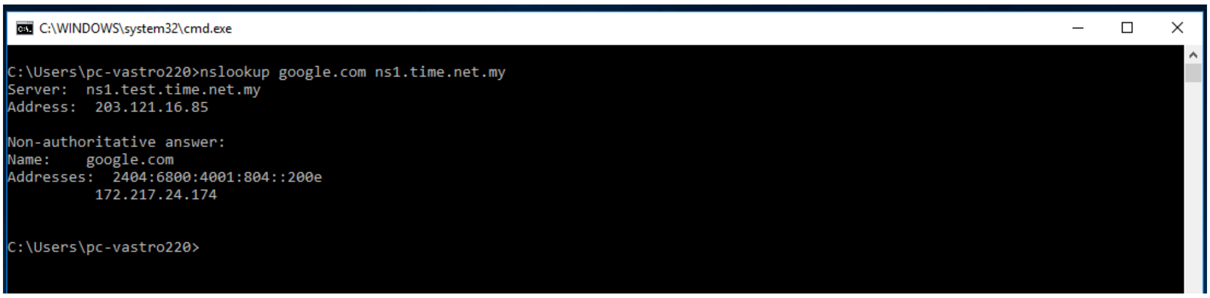
nslookup tool allows the host running the tool to query any specified DNS server for a DNS record. The queried DNS server can be a root DNS server, a top-level-domain DNS server, an authoritative DNS server, or an intermediate DNS server. To accomplish this task, nslookup sends a DNS query to the specified DNS server, receives a DNS reply from that same DNS server, and displays the result.

- To run it in Windows, open the Command Prompt (cmd) and run nslookup on the command line as shown in Figure C.1 and Figure C.2



```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup yahoo.com
Server:  ns1.utm.my
Address:  161.139.250.2

Non-authoritative answer:
Name:    yahoo.com
Addresses:  2001:4998:58:1836::11
          2001:4998:c:1023::4
          2001:4998:44:41d::4
          2001:4998:44:41d::3
          2001:4998:58:1836::10
          2001:4998:c:1023::5
          72.30.35.10
          98.137.246.7
          98.137.246.8
          98.138.219.231
          98.138.219.232
          72.30.35.9


C:\Users\pc-vastro220>nslookup -type=any yahoo.com
Server:  jb.utm.my
Address:  161.139.250.2

Non-authoritative answer:
yahoo.com       MX preference = 1, mail exchanger = mta5.am0.yahoodns.net
yahoo.com       MX preference = 1, mail exchanger = mta6.am0.yahoodns.net
yahoo.com       MX preference = 1, mail exchanger = mta7.am0.yahoodns.net
yahoo.com       AAAA IPv6 address = 2001:4998:c:1023::4
yahoo.com       AAAA IPv6 address = 2001:4998:44:41d::4
yahoo.com       AAAA IPv6 address = 2001:4998:44:41d::3
yahoo.com       AAAA IPv6 address = 2001:4998:58:1836::10
yahoo.com       AAAA IPv6 address = 2001:4998:c:1023::5
yahoo.com       AAAA IPv6 address = 2001:4998:58:1836::11
yahoo.com       nameserver = ns5.yahoo.com
yahoo.com       nameserver = ns2.yahoo.com
yahoo.com       nameserver = ns4.yahoo.com
yahoo.com       nameserver = ns1.yahoo.com
yahoo.com       nameserver = ns3.yahoo.com
yahoo.com       internet address = 98.137.246.7
yahoo.com       internet address = 98.137.246.8
yahoo.com       internet address = 98.138.219.231
yahoo.com       internet address = 98.138.219.232
yahoo.com       internet address = 72.30.35.9
yahoo.com       internet address = 72.30.35.10

C:\Users\pc-vastro220>
```

Figure C.1: nslookup result



```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup google.com ns1.time.net.my
Server:  ns1.test.time.net.my
Address:  203.121.16.85

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4001:804::200e
          172.217.24.174


C:\Users\pc-vastro220>
```

Figure C.2: nslookup result

1. Run nslookup to obtain the IP address of a www.microsoft.com server. What is the IP address of that server? Add screenshot to your answer.

```
C:\Users\USER>nslookup www.microsoft.com
Server:  ns3.utm.my
Address:  161.139.168.168

Non-authoritative answer:
Name:    e13678.dscb.akamaiedge.net
Addresses:  2600:1411:2000:380::356e
          2600:1411:2000:384::356e
          2600:1411:2000:390::356e
          2600:1411:2000:388::356e
          2600:1411:2000:385::356e
          104.83.197.169
Aliases:  www.microsoft.com
          www.microsoft.com-c-3.edgekey.net
          www.microsoft.com-c-3.edgekey.net.globalredir.akadns.net
```

IP Address: 104.83.197.169

2. Run nslookup to determine the non-authoritative DNS servers for domain microsoft.com. Add screenshot to your answer.

```
C:\Users\USER>nslookup www.microsoft.com
Server:  ns3.utm.my
Address:  161.139.168.168

Non-authoritative answer:
Name:    e13678.dscb.akamaiedge.net
Addresses:  2600:1411:2000:380::356e
          2600:1411:2000:384::356e
          2600:1411:2000:390::356e
          2600:1411:2000:388::356e
          2600:1411:2000:385::356e
          104.83.197.169
Aliases:  www.microsoft.com
          www.microsoft.com-c-3.edgekey.net
          www.microsoft.com-c-3.edgekey.net.globalredir.akadns.net
```

2.0 ipconfig

ipconfig can be used to show your current TCP/IP information, including your address, DNS server addresses, adapter type and so on.

- Information about host, use the following command: ipconfig /all

```
C:\WINDOWS\system32\cmd.exe
   Description . . . . . . . . . . . : TAP-ProtonVPN Windows Adapter V9
   Physical Address. . . . . . . . . : 00-FF-AA-18-73-0D
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . : utm.my
   Description . . . . . . . . . . . : Realtek PCIe GbE Family Controller
   Physical Address. . . . . . . . . : 00-24-E8-23-E3-3A
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::b8c5:7857:8b37:ddfe%4(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.60.82.216(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.252.0
   Lease Obtained. . . . . . . . . . : Tuesday, September 18, 2018 6:08:09 AM
   Lease Expires . . . . . . . . . . : Thursday, September 20, 2018 6:08:10 AM
   Default Gateway . . . . . . . . . : 10.60.80.1
   DHCP Server . . . . . . . . . . . : 10.60.80.1
   DHCPv6 IAID . . . . . . . . . . . : 33563880
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-1F-D8-2F-7E-00-24-E8-23-E3-3A
   DNS Servers . . . . . . . . . . . : 161.139.250.2
                                       161.139.168.168
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Ethernet 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : TAP-Windows Adapter V9
```

Figure C.3: ipconfig /all result

- ipconfig is also very useful for managing the DNS information stored in your host. Each entry shows the remaining Time to Live (TTL) in seconds.
  Command: ipconfig /displaydns

```
C:\WINDOWS\system32\cmd.exe
    Record Name . . . . . : wpad.utm.my
    Record Type . . . . . : 1
    Time To Live  . . . . : 5917
    Data Length . . . . . : 4
    Section . . . . . . . : Answer
    A (Host) Record . . . : 161.139.163.188


    aimsweb.utm.my
    ----------------------------------------
    Record Name . . . . . : aimsweb.utm.my
    Record Type . . . . . : 1
    Time To Live  . . . . : 16483
    Data Length . . . . . : 4
    Section . . . . . . . : Answer
    A (Host) Record . . . : 161.139.18.215


    my.utm.my
    ----------------------------------------
    Record Name . . . . . : my.utm.my
    Record Type . . . . . : 1
    Time To Live  . . . . : 16484
    Data Length . . . . . : 4
    Section . . . . . . . : Answer
    A (Host) Record . . . : 161.139.20.240


C:\Users\pc-vastro220>
```

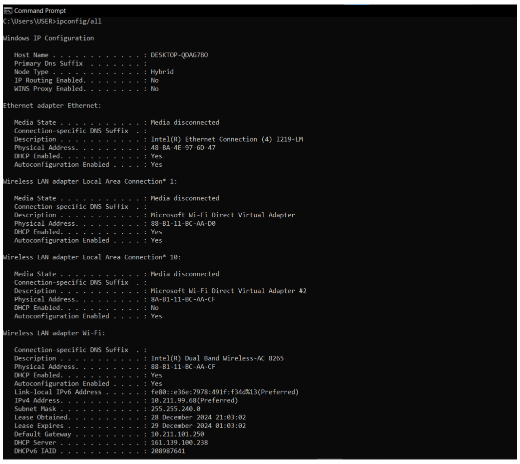
Figure C.4: ipconfig /displaydns result

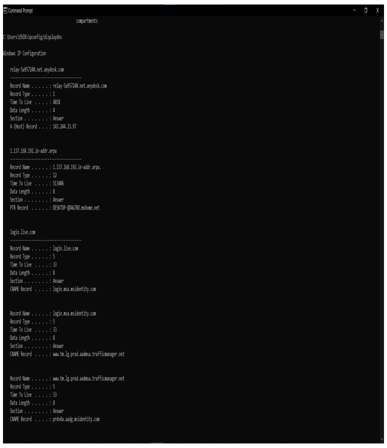
- Flushing the DNS cache clears all entries and reloads the entries from the hosts file.

  Command: ipconfig /flushdns

Figure C.5: ipconfig /flushdns result

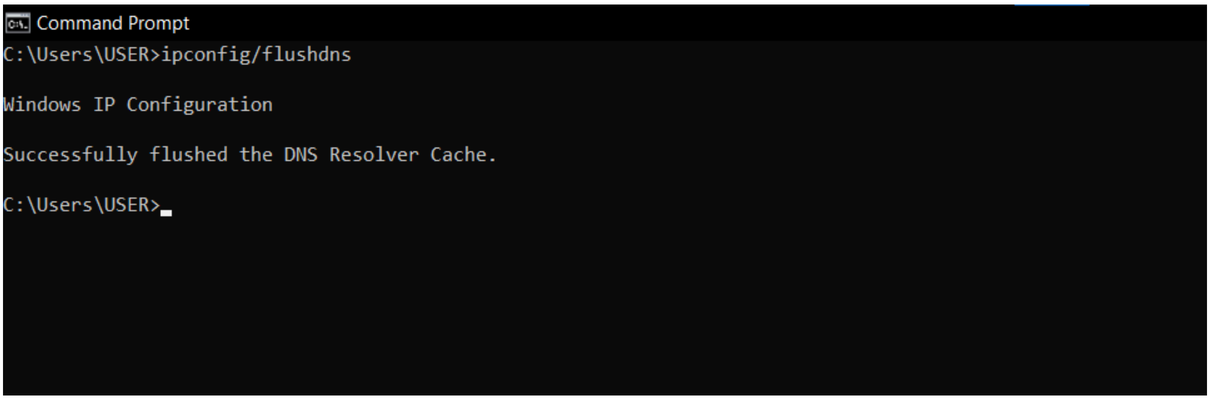
From my device





ipconfig/all

ipconfig/displaydns



ipconfig/flushdns

3.0 Tracing DNS with Wireshark

- Open packet trace file **dns-trace-1**. Answer the following questions.

1. Locate the DNS query and response messages. Are then sent over UDP or TCP? Add screenshots in your answer.

   Ans: UDP

```
Frame 8: 72 bytes on wire (576 bits), 72 bytes captured (576 bits)
Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.23
User Datagram Protocol, Src Port: 3163, Dst Port: 53
Domain Name System (query)
    Transaction ID: 0x006e
  ▸ Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
  ▸ Queries
    [Response In: 9]
```

DNS query message

```
Frame 9: 104 bytes on wire (832 bits), 104 bytes captured (832 bits)
Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM_10:60:99 (00:09:6b:10:60:99)
Internet Protocol Version 4, Src: 128.238.29.23, Dst: 128.238.38.160
User Datagram Protocol, Src Port: 53, Dst Port: 3163
Domain Name System (response)
    Transaction ID: 0x006e
  ▸ Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 2
    Authority RRs: 0
    Additional RRs: 0
  ▸ Queries
  ▸ Answers
    [Request In: 8]
    [Time: 0.000844000 seconds]
```

DNS response message

2. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

Ans:
Destination Port (query) : 53
Source Port (response) : 53

```
Frame 8: 72 bytes on wire (576 bits), 72 bytes captured (576 bits)
Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.23
User Datagram Protocol, Src Port: 3163, Dst Port: 53
    Source Port: 3163
    Destination Port: 53
    Length: 38
    Checksum: 0x8acb [unverified]
    [Checksum Status: Unverified]
    [Stream index: 1]
    [Stream Packet Number: 1]
  ▸ [Timestamps]
    UDP payload (30 bytes)
```

DNS query message

```
Frame 9: 104 bytes on wire (832 bits), 104 bytes captured (832 bits)
Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM_10:60:99 (00:09:6b:10:60:99)
Internet Protocol Version 4, Src: 128.238.29.23, Dst: 128.238.38.160
User Datagram Protocol, Src Port: 53, Dst Port: 3163
    Source Port: 53
    Destination Port: 3163
    Length: 70
    Checksum: 0xb0ba [unverified]
    [Checksum Status: Unverified]
    [Stream index: 1]
    [Stream Packet Number: 2]
  ▸ [Timestamps]
    UDP payload (62 bytes)
```

DNS response message

3. To what IP address is the DNS query message sent? Add screenshots in your answer.

Ans: 128.238.29.23

```
Frame 8: 72 bytes on wire (576 bits), 72 bytes captured (576 bits)
Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.23
   0100 .... = Version: 4
   .... 0101 = Header Length: 20 bytes (5)
 ▸ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
   Total Length: 58
   Identification: 0x229e (8862)
 ▸ 000. .... = Flags: 0x0
   ...0 0000 0000 0000 = Fragment Offset: 0
   Time to Live: 128
   Protocol: UDP (17)
   Header Checksum: 0xd281 [validation disabled]
   [Header checksum status: Unverified]
   Source Address: 128.238.38.160
   Destination Address: 128.238.29.23
   [Stream index: 1]
```

DNS query message

4. Examine the DNS query message. What "Type" of DNS query is it? Does the query message contain any "answers"? Add screenshots in your answer.

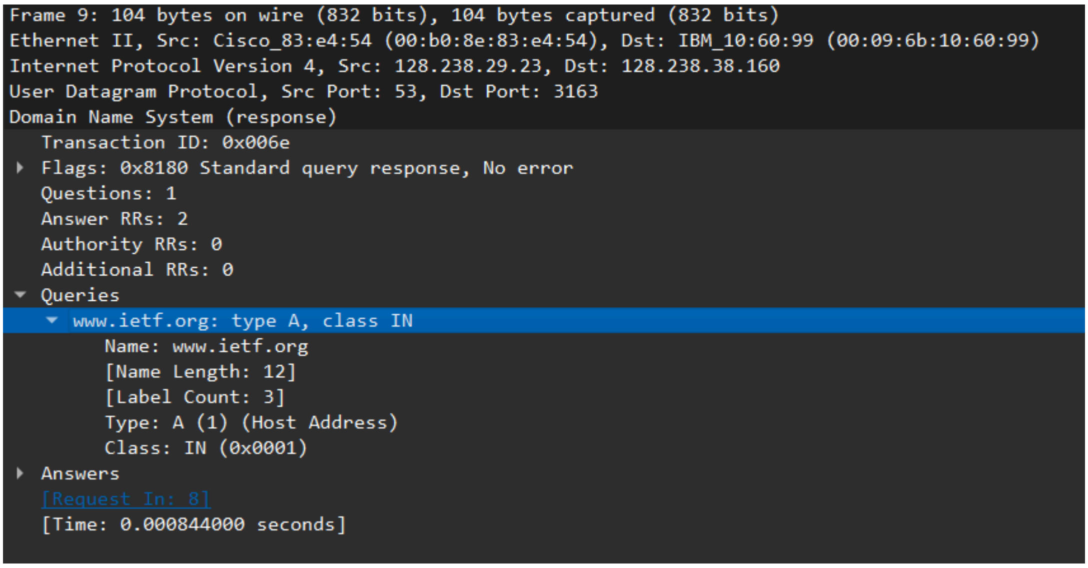
Ans: Type A. There's no answer in the query message.

```
Frame 8: 72 bytes on wire (576 bits), 72 bytes captured (576 bits)
Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.23
User Datagram Protocol, Src Port: 3163, Dst Port: 53
Domain Name System (query)
   Transaction ID: 0x006e
 ▸ Flags: 0x0100 Standard query
   Questions: 1
   Answer RRs: 0
   Authority RRs: 0
   Additional RRs: 0
 ▾ Queries
   ▾ www.ietf.org: type A, class IN
         Name: www.ietf.org
         [Name Length: 12]
         [Label Count: 3]
         Type: A (1) (Host Address)
         Class: IN (0x0001)
   [Response In: 9]
```

DNS query message

5. Examine the DNS response message. How many "answers" are provided? What do each of these answers contain? Add screenshots in your answer.

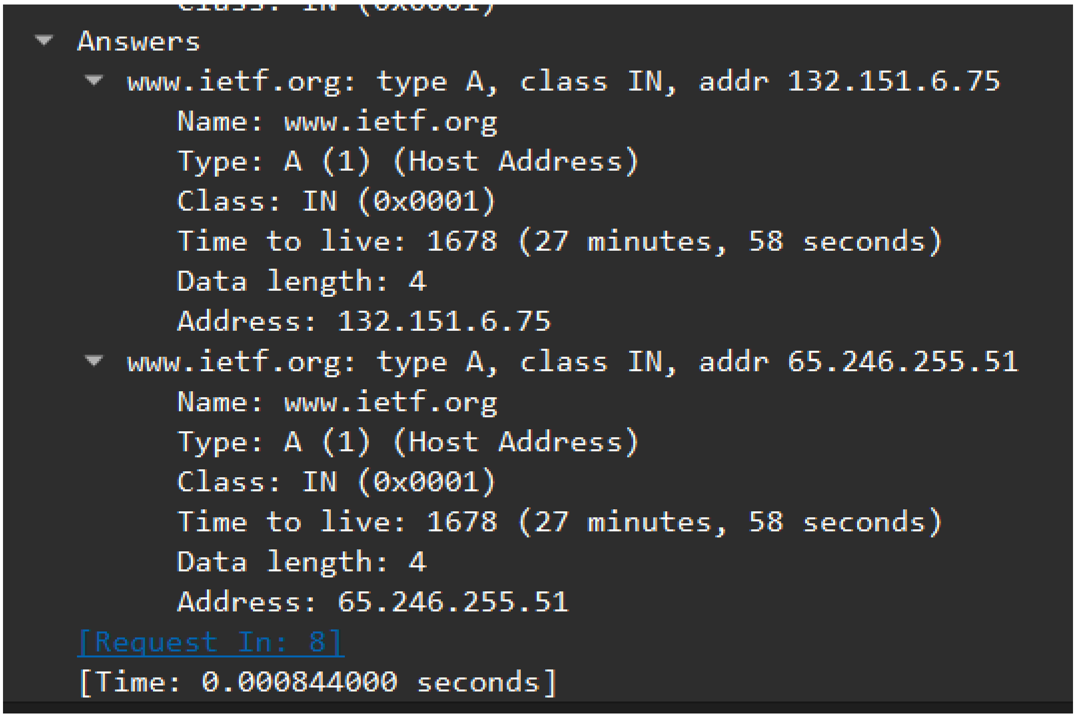
   Ans: There's 2 answers contained in the response message.



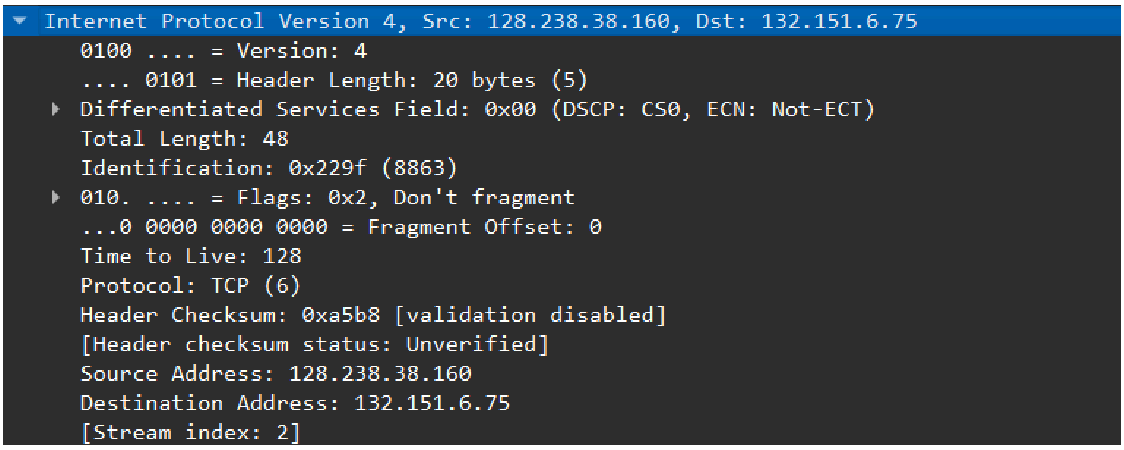
DNS response message

6. Consider the subsequent TCP SYN packet sent by your host. Does the destination IP address of the SYN packet correspond to any of the IP addresses provided in the DNS response message? Add screenshots in your answer.

Ans: Yes, the IP Address is 132.151.6.75



DNS Destination Response



TCP Destination Address

7. This web page contains images. Before retrieving each image, does your host issue new DNS queries?

Ans: No

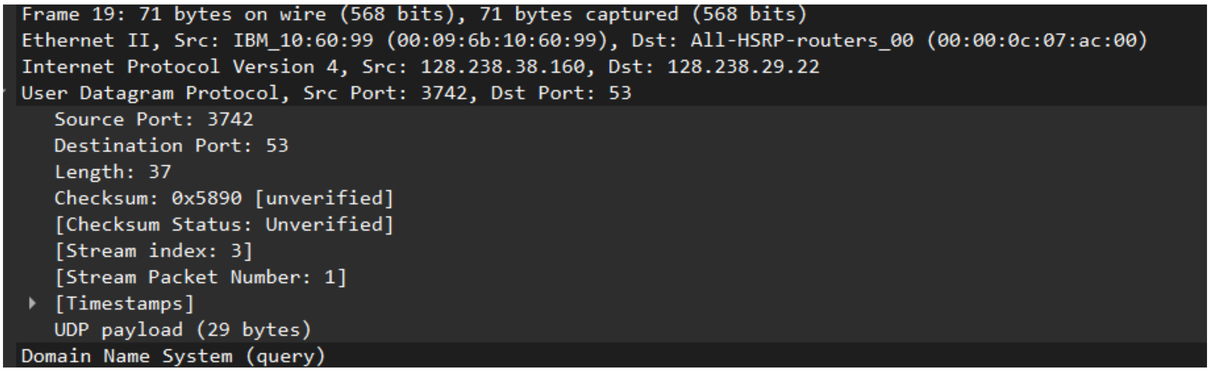
8	3.075845	128.238.38.160	128.238.29.23	DNS	72 Standard query 0x006e A www.ietf.org
9	3.076689	128.238.29.23	128.238.38.160	DNS	104 Standard query response 0x006e A www.ietf.org A 132.151.6.75 A 65.246.255.51
6	2.031956	128.238.38.2	224.0.0.2	HSRP	62 Hello (state Active)
83	4.604099	128.238.38.2	224.0.0.2	HSRP	62 Hello (state Active)
91	7.388092	128.238.38.2	224.0.0.2	HSRP	62 Hello (state Active)
13	3.096708	128.238.38.160	132.151.6.75	HTTP	429 GET / HTTP/1.1
20	3.153211	132.151.6.75	128.238.38.160	HTTP	1055 HTTP/1.1 200 OK (text/html)
28	3.191998	128.238.38.160	132.151.6.75	HTTP	320 GET /images/ietflogo2e.gif HTTP/1.1
31	3.192869	128.238.38.160	132.151.6.75	HTTP	314 GET /images/blue.gif HTTP/1.1
36	3.228451	132.151.6.75	128.238.38.160	HTTP	1212 HTTP/1.1 200 OK (GIF89a)
39	3.230578	132.151.6.75	128.238.38.160	HTTP	407 HTTP/1.1 200 OK (GIF89a)
52	3.286844	128.238.38.160	132.151.6.75	HTTP	317 GET /images/redstar.gif HTTP/1.1
53	3.287024	128.238.38.160	132.151.6.75	HTTP	319 GET /images/blue-line.jpg HTTP/1.1
55	3.309302	132.151.6.75	128.238.38.160	HTTP	1215 HTTP/1.1 200 OK (GIF89a)
61	3.321019	132.151.6.75	128.238.38.160	HTTP	741 HTTP/1.1 200 OK (JPEG JFIF image)
71	3.353822	128.238.38.160	132.151.6.75	HTTP	320 GET /images/isoc-small.gif HTTP/1.1
75	3.384003	132.151.6.75	128.238.38.160	HTTP	1269 HTTP/1.1 200 OK (GIF87a)

- Open packet trace file **dns-trace-2** for nslookup.
- We see from Wireshark that nslookup actually sent three DNS queries and received three DNS responses. For the purpose of this lab, ignore the first two sets of queries/responses, as they are specific to nslookup and are not normally generated by standard Internet applications. You should instead focus on the last query and response messages.
- Answer the following questions.

8. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

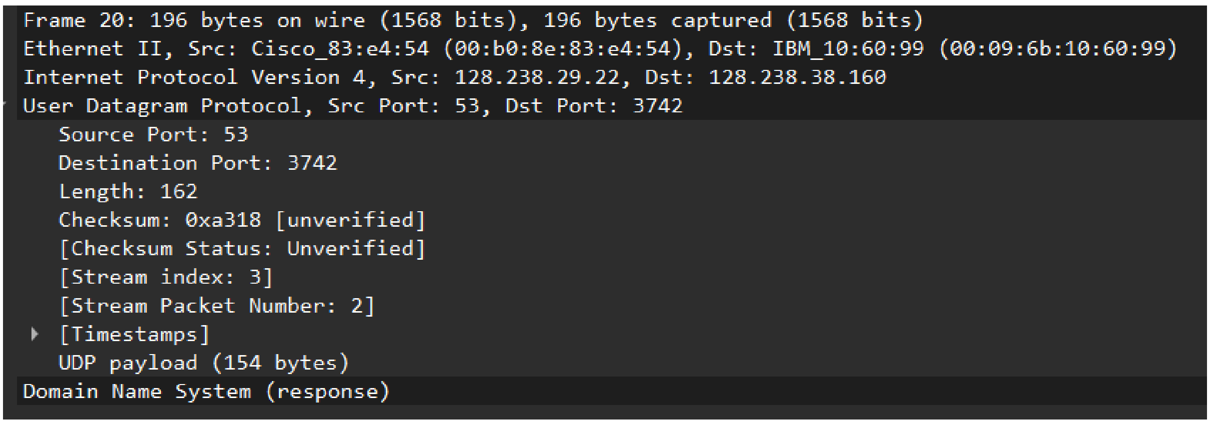
   Ans:

   Destination Port (query): 53
   Source Port (response): 53



```
Frame 19: 71 bytes on wire (568 bits), 71 bytes captured (568 bits)
Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.22
User Datagram Protocol, Src Port: 3742, Dst Port: 53
    Source Port: 3742
    Destination Port: 53
    Length: 37
    Checksum: 0x5890 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 3]
    [Stream Packet Number: 1]
  ▸ [Timestamps]
    UDP payload (29 bytes)
Domain Name System (query)
```

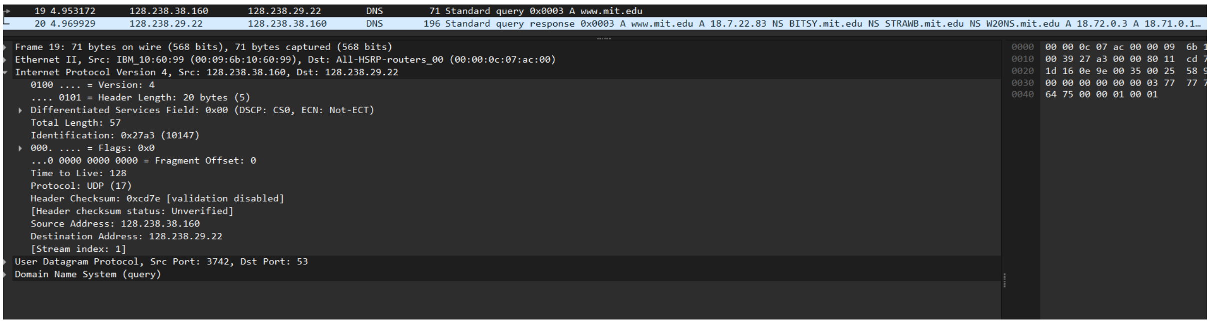
DNS Query Destination Port



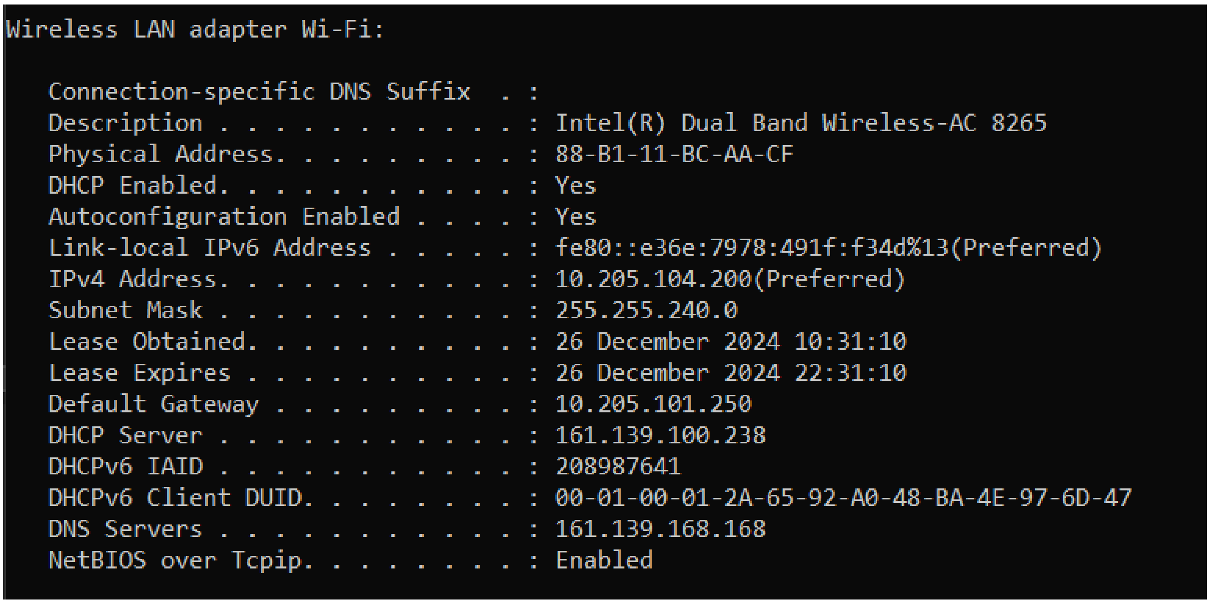
```
Frame 20: 196 bytes on wire (1568 bits), 196 bytes captured (1568 bits)
Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM_10:60:99 (00:09:6b:10:60:99)
Internet Protocol Version 4, Src: 128.238.29.22, Dst: 128.238.38.160
User Datagram Protocol, Src Port: 53, Dst Port: 3742
    Source Port: 53
    Destination Port: 3742
    Length: 162
    Checksum: 0xa318 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 3]
    [Stream Packet Number: 2]
  ▸ [Timestamps]
    UDP payload (154 bytes)
Domain Name System (response)
```

DNS Response Source Port

9. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server? Add screenshots in your answer.



Result of IP Address DNS Query message



Result of IP Address Ipconfig/all

Ans: The DNS Query message is sent to IP Address 128.238.29.22. This is not my default local DNS Server address.

10. Examine the DNS query message. What "Type" of DNS query is it? Does the query message contain any "answers"? Add screenshots in your answer.

    Ans: Type A, there's no answer in the query message.

```
Domain Name System (query)
    Transaction ID: 0x0003
  Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
  Queries
      www.mit.edu: type A, class IN
    [Response In: 20]
```

DNS query message

11. Examine the DNS response message. How many "answers" are provided? What do each of these answers contain? Add screenshots in your answer.

    Ans: There's 2 answers contained in the response message.

```
Domain Name System (response)
    Transaction ID: 0x0003
  Flags: 0x8580 Standard query response, No error
    Questions: 1
    Answer RRs: 1
    Authority RRs: 3
    Additional RRs: 3
  Queries
      www.mit.edu: type A, class IN
  Answers
      www.mit.edu: type A, class IN, addr 18.7.22.83
  Authoritative nameservers
      mit.edu: type NS, class IN, ns BITSY.mit.edu
      mit.edu: type NS, class IN, ns STRAWB.mit.edu
      mit.edu: type NS, class IN, ns W20NS.mit.edu
  Additional records
      BITSY.mit.edu: type A, class IN, addr 18.72.0.3
      STRAWB.mit.edu: type A, class IN, addr 18.71.0.151
      W20NS.mit.edu: type A, class IN, addr 18.70.0.160
    [Request In: 19]
    [Time: 0.016757000 seconds]
```

DNS response message